

EXAM MARKER

System Documentation & Functional Manual - V4

Laravel SaaS for AI-assisted paper correction, reporting and coaching-class management

Laravel V4

Platform

4

Plans

5

Primary Roles

PDF / DOC /

DOCX

Uploads

Document purpose	Developer handoff, system operation, user manual and QA reference
Audience	Owner, Laravel developer, QA tester, administrator and support team
User types	Platform Admin, Coaching Admin, Teacher, Student, Parent viewer
Plans	Free Mode, Mode 1, Mode 2, Mode 3
Core documents	QP - Question Paper; MS - Mark Scheme; WA - Written Answer
Status	Functional-system documentation aligned to V4 requirements

Core design rule: Plan permissions, user roles, organisation ownership and usage limits must be enforced at both UI and backend levels. Hidden buttons alone are not security.

1. Purpose and System Scope

Exam Marker is a multi-user Laravel web application that lets registered users upload an examination Question Paper (QP), an optional Mark Scheme (MS), and a Written Answer Paper (WA). The application processes the documents, produces AI-suggested question-wise marks and feedback, and then exposes different review, reporting and communication capabilities according to the user role and subscription plan.

- Public website for discovery, pricing, registration and login.
- Authenticated dashboards with role-aware and plan-aware navigation.
- Paper upload and queued checking workflow.
- Question-wise result review with confidence and feedback.
- Student record management and bulk workflows in higher plans.
- Parent data and communication in Mode 3 only.
- Platform administrator controls for plans, limits, accounts and usage.
- Private document storage and authenticated access.
- Demo marking adapter for testing and a replaceable production AI/OCR integration layer.

2. System Actors and Account Types

Actor	Purpose	Main capabilities	Key restrictions
Platform Admin	Operate entire SaaS	Manage plans, users, subscriptions, coaching accounts, limits, usage, support	Should not be constrained by customer plan features
Coaching Admin	Mode 3 organisation owner	Manage students, parent details, teachers, assessments, reports and communication	Limited by organisation plan and capacity
Teacher	Individual or organisation teacher	Create/check papers, review marks, manage permitted students	Mode 2 is single-teacher; Mode 3 may have multiple teachers
Student	Self-service user	Check own paper and view own results	Cannot change final marks or view other students
Parent viewer	Secure report recipient	Open expiring result link sent by coaching class	No general dashboard required; access only shared report

3. Plan Model and Feature Entitlements

Feature	Free	Mode 1	Mode 2	Mode 3
QP + optional MS + WA upload	Yes	Yes	Yes	Yes
Paper checks	1 lifetime	Plan limit	Plan limit	Plan limit
Question-wise marks	Yes	Yes	Yes	Yes
Total marks / basic feedback	Yes	Yes	Yes	Yes
Percentage / grade	No	No	Yes	Yes
Graphs / topic analysis	No	No	Yes	Yes
Strength & weakness analysis	No	No	Yes	Yes
Edit AI marks / feedback	No	No	Yes	Yes
Recheck selected question	No	No	Yes	Yes

Teacher final approval	No	No	Yes	Yes
Download result report	No	No	Yes	Yes
Download corrected paper	No	No	No	Yes
Student records	No	Limited	Yes	Yes
Bulk WA upload	No	No	Yes	Yes
Bulk student import	No	No	Yes	Yes
Parent name/email/mobile	No	No	No	Yes
Parent email / mobile report link	No	No	No	Yes
Communication history / delivery status	No	No	No	Yes
Multiple teachers	No	No	No	Yes
Teacher-wise paper allocation	No	No	No	Yes
Coaching administration	No	No	No	Yes

Important: Mode 2 may download the result/analysis report only. The corrected or annotated answer paper is a Mode 3 feature.

4. Public Website Modules

4.1 Homepage

- Explain the product in simple terms.
- Show QP/MS/WA workflow.
- Display four plans.
- Provide sample report/feature preview.
- Provide Register, Login and Check First Paper Free calls-to-action.
- No file upload should be available before registration and verification.

4.2 Pricing Page

Shows Free Mode, Mode 1, Mode 2 and Mode 3 side by side. Feature lists must come from the plan configuration or a shared feature definition so the pricing page cannot drift away from backend permissions.

4.3 Registration

- Full name
- Email
- Mobile
- Password and confirmation
- User type: Student, Teacher or Coaching Class
- Verification status
- Default Free Mode assignment after successful registration

4.4 Login and Password Recovery

On successful login the system redirects the user to the correct role-aware dashboard. Suspended users must be blocked. Password reset should use the configured Laravel mail channel.

5. Authentication, Authorisation and Dashboard Routing

Login identity	Resolved role	Plan context	Destination
Platform admin	admin	Platform scope	Admin Dashboard
Mode 3 coaching owner	coaching_admin	Mode 3 organisation	Coaching Dashboard
Teacher linked to coaching class	teacher	Organisation Mode 3	Teacher Workspace
Individual Mode 2 teacher	teacher	Mode 2 personal account	Teacher Dashboard
Student	student	Free / Mode 1 / assigned plan	Student Dashboard

- Every protected request must check authenticated user, ownership/organisation scope, role and plan permission.
- UI navigation should hide irrelevant modules, but routes/controllers must still enforce permissions.
- Coaching teachers may only see records within their coaching organisation and optionally only assigned assessments.
- Students may only see their own paper history and reports.

6. Main Sidebar Modules and How They Work

Sidebar item	Who sees it	Purpose	Primary actions
Dashboard	All authenticated roles	Snapshot of account and recent work	View usage, recent papers, plan status, alerts
Check New Paper	Student / Teacher / Coaching Admin	Start new assessment/check	Create assessment; upload QP, MS, WA; start checking
My Papers	Student / Teacher / Coaching Admin	Paper history and status	Search, filter, open, continue review, view result
Students	Mode 2 teacher / Mode 3 users	Student records	Add/edit students; bulk import in Mode 2/3; parent fields only Mode 3
Reports	Mode 2/3	Final result library	Open, print, download result report; corrected paper Mode 3 only
Analytics	Mode 2/3	Performance analysis	View averages, topics, strengths/weaknesses, question performance
Parent Messages	Mode 3	Parent communication	Send report, see delivery status, resend, open log
Teachers & Team	Mode 3 coaching admin	Teacher management	Add/suspend teachers, enforce limit, allocate work
Plan & Billing	Customer admins/users	Subscription and usage	Current plan, limits, upgrade, billing status
Settings	All authenticated users	Account preferences	Profile, organisation, password, notifications
Help & Support	All authenticated users	Support workflow	Create ticket, view ticket history
Notifications	All authenticated users	System alerts	Read/unread processing, plan, report and support alerts
Admin Plan Management	Platform admin	SaaS control plane	Plans, feature flags, limits, subscriptions, accounts
Logout	All authenticated users	End session	Invalidate session and return to public site

7. Check New Paper - End-to-End Functional Flow

- 1. Assessment setup:** User enters assessment title, subject, grade, board, date and total marks.
- 2. Upload QP + MS:** QP is compulsory. MS is optional. Both are presented on the same screen/page section. Accepted formats: PDF, DOC, DOCX.
- 3. Upload WA:** WA is compulsory. For batch workflows, files are mapped to student records.
- 4. Confirm:** Show selected filenames, student assignment, plan usage and warnings.
- 5. Queue job:** Create a processing job and move assessment to queued/processing status.
- 6. Document extraction:** Read text/layout and identify question structure. Production OCR/AI adapter is invoked here.
- 7. Marking:** Generate per-question maximum mark, AI-suggested mark, feedback, confidence and optional topic tags.
- 8. Review:** Mode 2/3 can manually review/edit/recheck. Lower plans see the permitted result only.
- 9. Finalise:** Teacher approval creates or updates final result and locks/version-controls the approved marks.
- 10. Report / parent communication:** Mode 2/3 can produce result report. Mode 3 may also download corrected paper and send parent communication.

8. QP, MS and WA Upload Module

Document	Required	Formats	Storage	Processing use
QP	Yes	PDF, DOC, DOCX	Private	Question extraction, marks available, structure
MS	No	PDF, DOC, DOCX	Private	Official marking criteria and mark points
WA	Yes	PDF, DOC, DOCX	Private	Student response to be assessed

- Validate extension and MIME type.
- Use configurable maximum file size.
- Reject images and unsupported formats.
- Keep uploads outside the public web root.
- Serve downloads/viewing through authorised controllers or signed URLs.
- When MS is absent, warn that the system is using a generated marking structure.

9. Paper Processing and AI Adapter

V4 may include a demo marker so the full workflow can be tested without API keys. Production deployment should replace or extend this adapter with OCR/document-understanding and AI-marking providers.

Stage	Input	Output	Failure handling
Extract QP	QP file	Question list, subquestions, marks, text	Set failed/retry; retain source file
Extract MS	MS file or none	Mark points / generated scheme	Continue with warning when optional MS absent
Extract WA	WA file	Student answer text/layout	Flag unreadable sections
AI evaluation	Question + criteria + student answer	Suggested mark, feedback, confidence, topic	Flag low-confidence result
Persist	Structured AI output	ai_check_results records	Transaction + validation
Notify	Job outcome	Dashboard notification/status	Retry notification delivery if needed

Production integration point: Keep AI/OCR provider code behind a service interface so providers can be replaced without rewriting controllers, views or database logic.

10. Question-wise Review Module

- Show original answer/paper context where available.
- Show maximum marks.
- Show AI-suggested marks.
- Show confidence: High / Medium / Low.
- Show AI feedback and criteria/mark points.
- Mode 2 and Mode 3: edit final mark and feedback.
- Mode 2 and Mode 3: recheck selected question.
- Record who edited a mark, old value, new value and timestamp.
- Final result must use teacher-final marks when present, otherwise the accepted AI mark.

Action	Free	Mode 1	Mode 2	Mode 3
View AI marks	Yes	Yes	Yes	Yes
Edit mark	No	No	Yes	Yes
Edit feedback	No	No	Yes	Yes
Recheck question	No	No	Yes	Yes
Teacher approve/finalise	No	No	Yes	Yes

11. Results, Reports and Downloads

- Result summary: total marks, obtained marks, percentage/grade when plan permits.
- Question-wise marks table.
- AI/teacher feedback.
- Topic analysis, strengths/weaknesses and graphs in Mode 2/3.
- Result report download in Mode 2 and Mode 3.

- Corrected answer paper download in Mode 3 only.
- Students should see only their own published/finalised reports.

11.1 Report states

- Draft - processing/review not complete.
- Review Required - AI confidence or teacher action pending.
- Finalised - approved result.
- Reported - report file/view generated.
- Shared - Mode 3 parent communication sent.

12. Students Module

Student records support repeated assessments, historical results and batch paper workflows. Data visibility is scoped to the owner or coaching organisation.

Field group	Mode 2	Mode 3
Student name / roll / grade / subject	Yes	Yes
Student login link	Optional	Optional
Bulk import	Yes	Yes
Parent name	No	Yes
Parent email	No	Yes
Parent mobile	No	Yes
Alternate contact	No	Yes

13. Bulk Upload and Bulk Import

- Mode 1: not available.
- Mode 2: bulk WA upload and bulk student import are available.
- Mode 3: bulk WA upload and bulk student import are available.
- Each uploaded WA must be mapped to exactly one student or placed in a reconciliation queue for manual assignment.
- Bulk import should validate duplicate email/roll number rules and show row-level errors before committing.

14. Analytics Module

- Class/teacher average and distribution.
- Question-wise success rate.
- Topic-wise performance.
- Strength and weakness analysis.
- High/low scoring questions.
- Trend across assessments when enough history exists.
- Mode 2: analysis limited to the individual teacher account.
- Mode 3: coaching admin sees organisation-level analytics; teachers may see their assigned scope.

15. Mode 3 Parent Management and Communication

- Parent details exist only in Mode 3 customer workflows.
- Email result report to parent.
- Send secure, expiring mobile/SMS report link through configured provider.
- Record communication channel, recipient, message/report reference, delivery state, sent timestamp and sender.
- Allow resend while preserving historical log.
- Secure parent links must be revocable and expire automatically.
- Parent viewer should not gain access to the full coaching dashboard.

Status	Meaning	Typical next step	Visible to	Audit?
queued	Waiting to send	Worker sends message	Mode 3 admin/teacher	Yes
sent	Provider accepted	Wait for delivery update	Mode 3	Yes
delivered	Confirmed delivery	No action	Mode 3	Yes
failed	Provider failure	Retry / correct contact	Mode 3	Yes
expired	Secure link expired	Regenerate if needed	Mode 3/parent	Yes

16. Mode 3 Teachers and Team

- Coaching admin can create teacher accounts up to the plan teacher limit.
- Teacher can be active/suspended without deleting historic work.
- Assessments/papers can be allocated to teachers.
- Teacher should only see organisation data allowed by assignment policy.
- Account suspension immediately blocks login while retaining audit/history.
- Mode 2 must never expose multiple-teacher management.

17. Plan and Billing Module

- Show current plan, start/expiry dates and account status.
- Show usage counters: paper checks, students, teachers, storage, email/SMS credits as applicable.
- Show upgrade options.
- Apply upgraded permissions after successful payment or admin assignment.
- On downgrade, preserve data but block unavailable actions.
- Expired plan behaviour should be configurable: read-only, grace period or upgrade wall.
- Failed payment must not delete customer data.

18. Settings Module

Area	User action	Notes
Profile	Edit name, mobile, profile basics	Email changes may require re-verification
Organisation	Mode 3 coaching name/address/branding	Branding optional feature
Password	Change password	Require current password
Notifications	Email/in-app preferences	Security-critical alerts may remain mandatory

Data	Request/delete owned data where policy allows	Respect audit/legal retention rules
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19. Notifications Module

- Paper queued/processing/completed/failed.
- Review required.
- Report finalised.
- Parent communication sent/failed.
- Plan nearing limit/expiry.
- Payment or subscription change.
- Support ticket reply.
- Notifications should support read/unread state and link to the relevant record.

20. Help and Support Module

- Authenticated user opens ticket with category, subject and description.
- System stores owner, status, priority and timestamps.
- Admin/support can respond and change status.
- User can view ticket history.
- Support module should not expose another customer's tickets.

21. Platform Admin Modules

Admin area	Functions	Critical controls
Users	Search, view, activate/suspend, assign role/plan	Prevent accidental self-lockout
Coaching classes	Create/assign plan/set limits/view usage	Single organisation; no multiple-branch feature
Plans & features	Edit feature flags and numeric limits	Database-driven; no hard-coded entitlements
Subscriptions	Assign/extend/expire plans	Audit all manual changes
Usage	Paper checks, storage, AI jobs, communication credits	Support cost/abuse monitoring
Support	Review tickets and failures	Escalation and notes
System	Queue failures, email failures, logs	Restricted to platform admins

22. Plan Feature Flag Architecture

Boolean permissions	Numeric limits
allow_percentage	paper_check_limit
allow_grade	student_limit
allow_graphs	teacher_limit
allow_topic_analysis	storage_limit
allow_manual_mark_editing	email_limit
allow_feedback_editing	sms_limit

allow_question_recheck	recheck_limit
allow_result_report_download	file_size_limit
allow_corrected_paper_download	assessment_limit
allow_bulk_answer_upload	retention_days
allow_bulk_student_import	
allow_student_records	
allow_parent_details	
allow_parent_email	
allow_parent_sms	
allow_multiple_teachers	
allow_teacher_assignment	
allow_custom_branding	
allow_student_portal	
allow_parent_portal	

Implementation: Controllers/services should call a common entitlement service such as `canFeature(user, feature)` and `remainingLimit(user, limit)` rather than duplicating plan logic.

23. Recommended Database Modules

Identity / billing	Academic / marking	Communication / operations
users	parents	reports
roles	assessments	parent_communications
permissions	assessment_students	email_logs
plans	document_uploads	sms_logs
plan_features	question_papers	usage_records
subscriptions	mark_schemes	support_tickets
payments	written_answers	notifications
coaching_classes	questions	audit_logs
coaching_class_users	mark_scheme_points	
students	ai_jobs	
	ai_check_results	
	teacher_reviews	
	final_results	

24. Core Status Model

Status	Meaning	Allowed transition examples
draft	Assessment exists; files incomplete	uploaded / cancelled
uploaded	Required files are present	queued
queued	Waiting for worker	processing / failed
processing	Extraction and marking running	review_required / completed / failed
review_required	Teacher action required	reviewed
reviewed	Teacher completed review	finalised
finalised	Approved marks locked/versioned	reported
reported	Report generated	sent_to_parent
sent_to_parent	Mode 3 communication completed	reported / resent
failed	Processing or delivery failed	queued / cancelled

25. Security and Data Protection

- Tenant/organisation scoping on every query.
- Laravel policies/gates/middleware for role and entitlement enforcement.
- Private storage for QP, MS, WA and reports.
- Signed/authorised download endpoints.
- Password hashing and CSRF protection.
- Rate limits on login, registration, uploads and job creation.
- Audit trail for mark edits, account changes, plan changes and parent communication.
- Sensitive parent contact details visible only to authorised Mode 3 users.
- No raw provider API keys in source control; use environment configuration.
- Validate Word/PDF files beyond filename extension.

26. Queue and Background Jobs

- Paper processing should run through Laravel queues for long-running OCR/AI work.
- Email and SMS/mobile delivery should be queued.
- Workers must be supervised in production.
- Failed jobs should be retrievable and visible to admin/support.
- Idempotency should prevent duplicate result generation or duplicate parent messages on retry.
- User-facing UI should poll or refresh status rather than hold a long HTTP request open.

27. Demo Engine vs Production Integrations

Capability	V4 demo behaviour	Production requirement
Handwriting/document OCR	May use demo/sample extraction	Connect OCR/document intelligence provider
AI marking	Demo result generator	Connect selected AI model/provider
Payments	Plan/admin assignment may be simulated	Connect payment gateway/webhooks
Email	Local/log/test mail possible	Configure transactional email provider
SMS/mobile links	UI/log may be present	Configure SMS/WhatsApp provider if required

28. User Workflows by Role

28.1 Student - Free Mode

1. Register and verify account.
2. Upload QP, optional MS and WA.
3. Use one lifetime check.
4. View question-wise marks, total marks and basic feedback.
5. Upgrade when free allowance is exhausted.

28.2 Mode 1 User

6. Login and start paper check.
7. Upload QP/MS/WA one paper at a time.
8. View marks, total and basic feedback.
9. Use paper history.
10. No bulk upload, graphs, downloads or parent features.

28.3 Mode 2 Teacher

11. Manage student records.
12. Create assessment and upload QP/MS.
13. Upload one or many WA files.
14. Review AI marks and confidence.
15. Edit/recheck selected questions.
16. Finalise result.
17. View analytics and download result/analysis report.
18. Cannot download corrected paper or manage parent contacts/multiple teachers.

28.4 Mode 3 Coaching Admin

19. Create/manage students including parent contact details.
20. Create teacher accounts within teacher limit.
21. Allocate papers/assessments to teachers.
22. Run same review/analytics workflow as Mode 2.
23. Download corrected answer papers.
24. Send report to parent by email or secure mobile link.
25. Monitor communication history and delivery status.

28.5 Platform Admin

26. Login to platform admin dashboard.
27. Manage users/coaching classes.
28. Assign or change plans and limits.
29. Manage plan feature flags.
30. Review usage, failed jobs, communications and support.
31. Suspend/activate accounts while retaining history.

29. Functional Acceptance Checklist

- ☐ 1. Unregistered users cannot upload papers.
- ☐ 2. Only PDF, DOC and DOCX are accepted.
- ☐ 3. QP and WA are mandatory; MS is optional.
- ☐ 4. Free Mode allows one lifetime check.
- ☐ 5. Mode 1 has no bulk upload.
- ☐ 6. Mode 2 supports bulk WA and bulk students.
- ☐ 7. Mode 2 supports manual edit, recheck, analytics and result-report download.
- ☐ 8. Mode 2 cannot download corrected paper.
- ☐ 9. Mode 2 cannot store parent contact details or create multiple teachers.
- ☐ 10. Mode 3 supports parent contact details, communication and multiple teachers.
- ☐ 11. Mode 3 supports corrected-paper download.
- ☐ 12. All customer data is tenant-scoped.
- ☐ 13. All plan restrictions are enforced in backend, not just the UI.
- ☐ 14. AI marks are shown as suggested until accepted/finalised.
- ☐ 15. All mark edits and key administrative changes are audited.
- ☐ 16. No Mode 3 Plus and no multiple-branch module exists.

30. Developer Handoff - Recommended Code Organisation

- Controllers: keep request orchestration thin.
- Services: entitlement/plan service, document service, paper marker service, report service, communication service.
- Policies: assessment, student, report, coaching class and admin resources.
- Jobs: ProcessPaperJob, GenerateReportJob, SendParentReportJob.
- Events/notifications: PaperCompleted, ReviewRequired, PlanLimitReached, ParentReportSent.
- Repositories/query scopes are optional, but tenant scoping must be central and consistent.
- Feature tests should cover every role-plan combination for protected routes.

31. Deployment and Operations Checklist

- Configure APP_KEY and production environment.

- Configure MySQL or supported SQLite environment.
- Set private storage and symbolic/public links only where appropriate.
- Configure queue driver and run workers.
- Configure scheduler/cron if needed.
- Configure mail provider.
- Configure AI/OCR/payment/SMS providers when moving beyond demo mode.
- Run migrations and seed only intended production data.
- Set backups for database and private document storage.
- Enable HTTPS, secure cookies and production logging.
- Configure monitoring for queue failures and provider errors.

32. Summary

Exam Marker V4 should be treated as one coherent permission-driven platform rather than separate websites for each plan. Authentication identifies the user, role determines what type of work the user can perform, plan entitlements determine which commercial features are available, and organisation ownership determines which records the user can access. The same assessment and marking engine is reused across Free, Mode 1, Mode 2 and Mode 3, while higher plans progressively unlock analytics, editing, reporting, parent communication and team management.

Final scope: Four plans only: Free, Mode 1, Mode 2 and Mode 3. Mode 2 has no corrected-paper download, parent contacts or multiple teachers. Mode 3 adds those capabilities. No Mode 3 Plus and no multiple-branch feature.